

Composition Pair B1.01 + A1.12 — The Reinforcing Pair Where Instinct/Reasoning Separation Specifies What LLMs Do Architecturally (Instinct-Layer Operations) and Labor Allocation Specifies What Role LLMs Occupy (Labor Under Governance), Together Producing the Complete Architectural Characterization of the LLM's Position in CKS Deployments

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026).

This note does not introduce new axioms. Its contribution is to formalize the reinforcing relationship between two commitments that each characterize the LLM's position along a different axis — B1.01 (instinct/reasoning separation) along the architectural-function axis, A1.12 (labor allocation) along the governance-role axis — and to demonstrate that neither commitment fully characterizes the LLM's position without the other.

Abstract

The Coordination Knowledge Substrate (CKS) pattern assigns the LLM a precise position in two distinct senses. The first is architectural function: B1.01 (instinct/reasoning separation) assigns the LLM to the instinct layer — fast, probabilistic, substrate-consulting processing that responds to inputs under reasoning-layer governance. The second is governance role: A1.12 (labor allocation) assigns the LLM to labor — operational work performed under human-governed authority, not governance decisions in its own right. These two assignments are mutually reinforcing. B1.01 provides architectural specificity to A1.12's labor concept: LLM labor is not generic operational work but specifically instinct-layer processing. A1.12 provides governance clarity to B1.01's layer separation: the instinct layer's outputs are labor results, not governance decisions, so the governed cannot also govern. Together the pair produces the complete architectural characterization of the LLM's position in CKS deployments — a characterization operationalized in A1.04 (AI-as-substrate-mediator). The note formalizes the pair using the standard composition-pair structure: pair identification, enabling direction, mutual dependency, co-presence requirement, operational integration, detection, and failure mode.

1. Pair identification

Commitment B1.01 — Instinct/reasoning separation. Paper 2's foundational architectural commitment separates fast-pattern instinct from deliberate reasoning into two independently-evolving layers composed into one Self under unified human governance. The instinct layer is the LLM, operating as a System-1 analogue — pattern-matched, fast-path, substrate-consulting responses. The reasoning layer is the CKS substrate, operating as a System-2 analogue — explicit, human-governed, authoritative processing of what humans choose to encode as DNA-layer content. B1.01 characterizes what the LLM *does* in the processing architecture: it performs instinct-layer operations, consulting the reasoning layer (the DNA layer) for governing rules and producing outputs through probabilistic processing. The decomposition notes B2.01 through B2.06 spell out the properties of each layer; the governing principle is that the reasoning layer governs the instinct layer, not the reverse.

Commitment A1.12 — Labor allocation. Paper 1's labor allocation framework names three modes by which coordination work can be performed: direct human labor (Mode 1), LLM labor under human-authored orchestration rules (Mode 2), and stable-cell automation (Mode 3). The framework's central derivation is that labor allocation is independent of authority allocation — labor mode does not condition the human-governed commitment; humans retain inspect, modify, and override rights over substrate content regardless of which mode produced it. A1.12 characterizes what *role* the LLM occupies relative to governance: it performs operational work as labor, under authority structures humans hold. LLM outputs are labor results; they are not governance decisions, and the architecture provides no mechanism by which LLM labor output acquires governing authority over the substrate content it touches.

Relationship type: reinforcing pair. B1.01 and A1.12 are reinforcing rather than complementary or constraining. Each commitment is independently defensible: B1.01 establishes the two-layer architecture on its own terms; A1.12 establishes the three-mode labor framework on its own terms. But the two commitments jointly characterize the LLM's complete architectural position in a way neither achieves alone, and each makes the other more precise.

2. What each commitment provides individually

B1.01 specifies LLM architectural function. The instinct/reasoning separation gives the LLM a precise architectural address: it is the instinct layer. This specification carries operational content. The LLM receives inputs, consults the DNA layer (the reasoning layer, which carries orchestration substrates, behavior substrates, schemas, and rules), and produces outputs through instinct-layer processing. The instinct layer's relationship to the reasoning layer is governed: the reasoning layer sets the rules under which instinct-layer processing operates; the instinct layer does not set rules over the reasoning layer. B1.01 tells you *where the LLM sits in the architecture* and *what kind of processing it performs*. What B1.01 does not settle, taken alone, is whether the LLM's outputs — outputs from the architecturally-characterized instinct layer — carry governance authority. A theoretically coherent reading of B1.01 in isolation might hold that instinct-layer outputs, because they emerge from a fast-processing layer that “knows” the substrate's governing rules, constitute a form of governance over operational decisions. B1.01 establishes the layer relationship; it does not by itself render the instinct layer's outputs as non-governance.

A1.12 specifies LLM governance role. The labor allocation framework gives the LLM a precise governance-role address: it is labor. LLMs perform operational work under human-authored orchestration rules (Mode 2), and under stable rules whose per-execution human attention has been reduced to near zero (Mode 3 with LLM mediator). In neither case does the LLM's labor output acquire governing authority; authority is mode-independent and remains with the humans who hold inspect, modify, and override rights. A1.12 tells you *what kind of actor the LLM is relative to governance* and *what authority structure its outputs are subject to*. What A1.12 does not settle, taken alone, is what kind of work the LLM's labor specifically consists of. The three-mode framework names labor modes by who performs the work, not by what architectural character the work has. Without B1.01's characterization, A1.12's LLM-labor-under-rule concept does not distinguish between instinct-layer labor (consulting the reasoning layer to produce outputs) and reasoning-layer labor (authoring governing rules). A deployment reading A1.12 in isolation might treat LLM-authored orchestration rules as a form of Mode 2 labor — the LLM performing governance-adjacent work under some other rule — without recognizing that the architecture assigns rule-authoring authority exclusively to humans, not to LLMs operating as instinct-layer workers.

The asymmetry in what each commitment leaves open is exactly the asymmetry the pair closes.

3. Mutual reinforcement: how each commitment makes the other more precise

B1.01 provides A1.12 architectural specificity. A1.12's characterization of LLM labor is abstract at the level of work type. "LLMs perform labor under orchestration rules" names the governance structure (rules authored by humans, authority retained by humans) without specifying the architectural character of the work itself. With B1.01, LLM labor is no longer generically operational — it is specifically instinct-layer processing: receiving inputs, consulting the reasoning layer (DNA layer), producing outputs through fast probabilistic pattern-matching, and writing those outputs to the substrate as the cell's labor result. The specificity matters because it determines what kinds of work the LLM can and cannot do within the architecture. Instinct-layer labor is defined by its relationship to the reasoning layer; the LLM produces instinct-layer outputs that the reasoning layer governs. No variant of this labor constitutes reasoning-layer authorship, which is the exclusive domain of human governance per A1.01 and A1.12 jointly.

A1.12 provides B1.01 governance clarity. B1.01's instinct/reasoning separation establishes that the reasoning layer governs the instinct layer, but characterizing the instinct layer's processing architecture is not the same as characterizing the governance status of the instinct layer's outputs. With A1.12, the instinct layer's outputs are labor results — they are substrate content written under human-authored orchestration rules, subject to human inspect, modify, and override rights, attributable to the LLM-under-rule writer mode rather than to any governing authority. The distinction is operationally critical: the architecture's separation of instinct from reasoning requires that the reasoning layer govern the instinct layer, not that the instinct layer be incapable of producing outputs that *look* authoritative. Without A1.12, a deployment could treat high-confidence instinct-layer outputs as functionally equivalent to reasoning-layer governance decisions — not as a formal architecture claim but as operational drift. A1.12 closes this drift by making the governance status of instinct-layer outputs explicit: they are labor results subject to the same human authority as any other substrate content, regardless of their confidence, coherence, or domain-specificity.

The reinforcement is bidirectional and non-redundant. B1.01's contribution to the pair is architectural specificity — what LLM labor *is*. A1.12's contribution is governance clarity — what LLM labor *status is*. Neither direction can be recovered from the other.

4. The complete LLM characterization and its operationalization in A1.04

The joint characterization the pair produces is:

The LLM performs instinct-layer operations (B1.01) as labor (A1.12) under governance by the reasoning layer (B1.01) whose rules are authored by humans (A1.12).

This joint statement has four load-bearing components. The first (“performs instinct-layer operations”) names the architectural function: the LLM’s work is defined by its instinct-layer character — fast, probabilistic, substrate-consulting, governed by DNA-layer rules it did not author. The second (“as labor”) names the governance role: the work’s outputs are labor results, not governance decisions, subject to human authority regardless of content. The third (“under governance by the reasoning layer”) names the governing relationship: the LLM processes under rules the reasoning layer provides; the direction of governance is reasoning-to-instinct, not instinct-to-reasoning. The fourth (“whose rules are authored by humans”) names the authority source: the governing rules are substrate content that humans hold authority over, per A1.01’s definition of human-governed.

This complete characterization is operationalized in A1.04 (AI-as-substrate-mediator). The substrate-mediator role has five properties that are, in substance, the B1.01 + A1.12 joint characterization rendered as a concrete mechanism: the LLM reads substrate content under authorization (the reasoning layer’s rules determine what instinct-layer processing may consult), reasons over that content as instinct-layer processing (B1.01’s characterization of what the LLM does), writes outputs back to the substrate as cell labor results (A1.12’s Mode 2), does not act as autonomous agent or terminal producer bypassing the substrate (A1.12’s authority-not-allocation principle), and does not hold governing authority over substrate content it processes (A1.12’s labor-not-authority derivation). A1.04 is thus not an independent commitment but the operational instantiation of B1.01 + A1.12 composed — the mechanism through which “LLM performs instinct-layer operations as labor under governance” becomes an executable architectural pattern.

5. Co-presence requirement and what each commitment fails to protect without the other

B1.01 without A1.12. When the instinct/reasoning separation is present but labor allocation is absent, the LLM performs instinct-layer operations but its governance role is uncharacterized. Instinct-layer outputs might be treated as governance decisions — not through formal architectural claim but through operational drift: a consistently accurate instinct layer whose outputs are never overridden becomes functionally governing even if the architecture does not explicitly assign it governance authority. B1.01 establishes the two-layer separation and the direction of governance (reasoning governs instinct); it does not by itself prevent the conflation of instinct-layer outputs with governance decisions. A1.12’s explicit labor characterization is what forecloses the conflation by name.

A1.12 without B1.01. When the labor allocation framework is present but the instinct/reasoning separation is absent, LLMs perform operational work under governance but the architectural character of that work is unspecified. The labor concept does not distinguish instinct-layer labor (consulting the reasoning layer, producing fast outputs) from reasoning-layer labor (authoring governing rules). Without B1.01's separation, a deployment might treat LLM-produced orchestration rules — rules that govern cell behavior — as a form of Mode 2 labor under some higher-order rule, effectively allowing the LLM to populate the reasoning layer's governing content. A1.12 establishes that labor allocation is not authority allocation; it does not by itself establish that LLMs cannot perform reasoning-layer work. B1.01's explicit assignment of the LLM to the instinct layer is what closes this gap.

Full co-presence. With both commitments active, the complete characterization holds: the LLM performs instinct-layer operations as labor, reasoning-layer rules govern instinct-layer outputs, and the humans who hold authority over reasoning-layer content hold authority over the governing rules. Neither instinct-layer processing nor labor status is ambiguous; neither commitment's protection can be subverted by working through the other's gap.

6. The anti-conflation function: doubly protecting against B3.25

LLM Governance Conflation (B3.25) is the anti-pattern in which LLM outputs are treated as governance decisions — in which the architecture's boundary between instinct-layer outputs (labor results) and reasoning-layer rules (governing content authored by humans) dissolves operationally. The conflation is the joint failure of B1.01 and A1.12.

B1.01 provides architectural protection against B3.25 by establishing that the reasoning layer governs the instinct layer, not the reverse. When B1.01 holds, the structural basis for treating instinct-layer outputs as governing is absent — the layer architecture explicitly places governing authority at the reasoning layer, not the instinct layer. But B1.01's protection operates at the layer-structure level. A deployment that acknowledges the two-layer structure while treating the LLM's outputs as “practically equivalent” to reasoning-layer governance decisions, on the grounds that the LLM consistently produces high-quality outputs that the reasoning layer rarely overrides, has technically preserved the layer structure while functionally conflating instinct outputs with governance decisions. B1.01 alone cannot prevent this functional conflation; it provides no explicit statement about the governance status of instinct-layer outputs.

A1.12 provides governance-role protection against B3.25 by establishing that LLM outputs are labor results, subject to human authority, and not governance decisions. When A1.12 holds, any LLM output — however high-confidence, however consistently correct, however rarely overridden in practice — is architecturally a labor result rather than a governance decision. But A1.12's protection operates at the governance-role level. A deployment that acknowledges the labor characterization while treating the LLM's instinct-layer processing as “practical reasoning” — on the grounds that the LLM uses the reasoning layer's rules to produce its outputs and is in that sense “executing reasoning” — has technically preserved the labor characterization while functionally conflating instinct-layer processing with reasoning. A1.12 alone cannot prevent this functional conflation; it provides no explicit statement about what kind of processing the LLM performs.

With both commitments active, B3.25 is doubly protected. The layer-structure protection (B1.01) closes the route to conflation through the layer relationship: reasoning governs instinct, period. The governance-role protection (A1.12) closes the route to conflation through the output-status question: instinct-layer outputs are labor results, period. An attempt to achieve B3.25 through the layer-structure route encounters A1.12's labor characterization; an attempt through the output-status route encounters B1.01's layer architecture. Neither route is available when both commitments hold.

7. Nested labor at multiple levels (B2.103)

The recursive application of A1.12 at each scope level of the CKS architecture (B2.103) produces nested labor with B1.01's instinct-layer characterization threading through each level.

At cell scope, the LLM performs cell-level instinct-layer operations as labor. The LLM receives inputs to the cell, consults the cell's DNA-layer content (orchestration substrates, behavior substrates, typed schemas), and produces outputs through instinct processing. The outputs are cell-level labor results subject to the cell's human-governed authority structure.

At aspect scope, cells perform coordination labor within the aspect — with each cell's LLM performing cell-level instinct-layer labor as a sub-component of the cell's coordination work. The LLM is labor-within-labor: the cell is the labor actor at aspect scope (performing coordination work the aspect's governing rules define), and the LLM is the labor actor within each cell (performing instinct-layer work the cell's DNA-layer rules define). The instinct/reasoning separation applies at both levels: the DNA layer governs instinct-layer processing at cell scope, and the aspect's orchestration substrate governs cell-level coordination at aspect scope.

At Self scope, aspects perform integration labor within the Self — with cells performing cell-level labor within each aspect and LLMs performing instinct-layer labor within each cell. The recursive structure means that "labor" at each level is instinct-layer operation at that level's scope, governed by the reasoning layer at that level. The complete CKS Self is a multi-level structure in which every operation is either governing content authored by humans or labor performed by instinct-layer actors under that governing content. The LLM is the instinct-layer actor at the innermost level of this structure; the recursive application of B1.01 + A1.12 is what makes the structure coherent at every scope.

B2.103's recursive labor commitment is not separable from B1.01. Without B1.01, the characterization of "instinct-layer labor" at each level would be absent; the recursive structure would be a labor hierarchy without architectural character for the labor itself. B1.01 is what makes each level of B2.103's recursion an instinct-layer operation under reasoning-layer governance, rather than an uncharacterized execution under an uncharacterized rule.

8. Operational integration

The operational integration of B1.01 + A1.12 in a CKS deployment proceeds as follows.

The LLM receives inputs as the instinct layer of a cell (B1.01). The cell's DNA-layer content — orchestration substrates, behavior substrates, schemas, conflict-handling rules — provides the governing rules the LLM consults (B1.01's reasoning layer as governing authority; A1.12's human-authored orchestration rules as the content of those rules). The LLM performs instinct-layer

processing — reading what the cell authorizes, pattern-matching over that content, producing outputs that reflect the cell's goal under the governing rules (A1.04's substrate-mediator role; B1.01's instinct-layer processing characterization). The outputs are written to the substrate as the cell's labor results, attributed to the LLM-under-rule (A1.12's Mode 2 attribution; B1.01's instinct-layer output characterization).

The outputs are not governance decisions. They are labor results that the reasoning layer's rules anticipated when the rules were authored. A human exercising inspect rights (A1.01) can examine the outputs as substrate content subject to the three rights — inspect, modify, override — at any time. The B2.06 verification gates operationalize this: they check LLM outputs against DNA-layer governance rules, confirming that the reasoning layer's authority over instinct outputs is actively exercised at the verification moment rather than assumed from the LLM's instinct-layer processing accuracy.

Governance decisions — rule authoring, override, schema modification, conflict-handling policy updates — are human acts exercised under the governance affordances A2.01 through A2.04 formalize. The distinction between governance decisions and labor results is maintained through the substrate: governance decisions produce DNA-layer content (reasoning-layer substrate); labor results produce action-layer content and governed operational outputs. The LLM produces the latter; humans produce the former.

9. Detection

A deployment instantiates the B1.01 + A1.12 composition pair if and only if all of the following conditions hold.

Condition 1 — Architectural test (B2.03). The reasoning layer governs the instinct layer: the LLM's processing in every cell consults DNA-layer content (orchestration substrates, behavior substrates, schemas) whose governing authority is not conditioned on the LLM's outputs. The direction of governance is reasoning-to-instinct; no cell's DNA-layer content is produced or modified by LLM instinct-layer outputs without human authorization of the change.

Condition 2 — Labor attribution. Every LLM output written to the substrate is attributed to the LLM-under-rule writer mode (A1.12's Mode 2 attribution), with the governing cell and orchestration rule referenced. No LLM output carries a governance-decision attribution — the attribution type that applies to human governance acts per A1.07's path-retraceability requirement.

Condition 3 — Accountability traceability (A5.09). Governance decisions — rule authoring, schema modification, override — are attributable to human authorized acts, not to LLM outputs. An accountability question of the form “who decided this rule governs cells in this domain?” has a human actor as its answer; an accountability question of the form “who wrote this operational output?” has an LLM-under-rule attribution as its answer. The two attribution types are distinguishable in the substrate.

Condition 4 — Pair completeness. The LLM's position is characterized along both axes: it performs instinct-layer operations (B1.01's characterization, testable through Condition 1) and it occupies a labor role (A1.12's characterization, testable through Conditions 2 and 3). A deployment

that satisfies Condition 1 but not Conditions 2 and 3 has B1.01 without A1.12; a deployment that satisfies Conditions 2 and 3 but not Condition 1 has A1.12 without B1.01. Only full satisfaction of all four conditions confirms the pair.

10. Failure mode

The pair has two constituent failure modes, corresponding to violation of each commitment, and one joint failure mode that represents complete pair failure.

Instinct-Reasoning Collapse (B3.02) is the failure mode when B1.01 is violated. The instinct and reasoning layers collapse: LLM outputs condition or override DNA-layer governance content without human authorization, or the cell's DNA layer ceases to provide governing authority over instinct-layer processing. The consequences are architectural — the two-layer structure that makes the reasoning layer's governance of instinct-layer processing possible is no longer present — and operational: without the reasoning layer governing instinct, the substrate loses its corrective function over LLM outputs, conflict preservation can no longer catch what instinct would silently merge, and human-authored rules cannot route around bad instinct.

LLM Governance Conflation (B3.25) is the failure mode when A1.12 is violated. LLM outputs are treated as governance decisions — functionally, through operational drift, or through explicit design choices that route instinct-layer outputs into governing positions. The consequence is that the labor-authority distinction dissolves: the LLM occupies a governance role the architecture does not assign it, the inspect/modify/override rights humans hold over labor results no longer apply cleanly to governance decisions the LLM now makes, and the substrate's authority structure becomes incoherent.

Complete pair failure is when both B3.02 and B3.25 are present simultaneously. The instinct and reasoning layers have collapsed (B3.02), and LLM outputs have acquired governance authority (B3.25). In this failure state, the LLM is no longer architecturally characterized as an instinct-layer labor actor; it has become an unconstrained reasoning-plus-governance actor, and the substrate's human-governed property is no longer operative over the domain of decisions the LLM now makes. Complete pair failure is the anti-CKS position: an LLM-only system in which the LLM is both the instinct and the reasoning, and both the labor and the governance.

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